ONE MILLION HEART CELLS

INTEGRATING PUBLIC DATA ON GALAXY TO EXPLORE HOW CARDIOMYOCYTES CHANGE IN ATRIAL FIBRILLATION

Marisa Loach, Daniel M. Johnson, Wendi Bacon
School of Life, Health and Chemical Sciences, The Open University, UK

CARDIOMYOCYTES & AF

Atrial Fibrillation (AF) is the most common form of arrhythmia and a contributing factor for 1 in 5 strokes. AF occurs when cells outside the natural pacemaker start producing electrical signals that cause the surrounding muscle to contract at the wrong time. While cardiomyocytes have been implicated in AF, difficulties in isolating and analysing them at the single cell level have limited our understanding.

- too FRAGILE to isolate
- too BIG to capture
- too many mitochondria - often filtered out during analysis

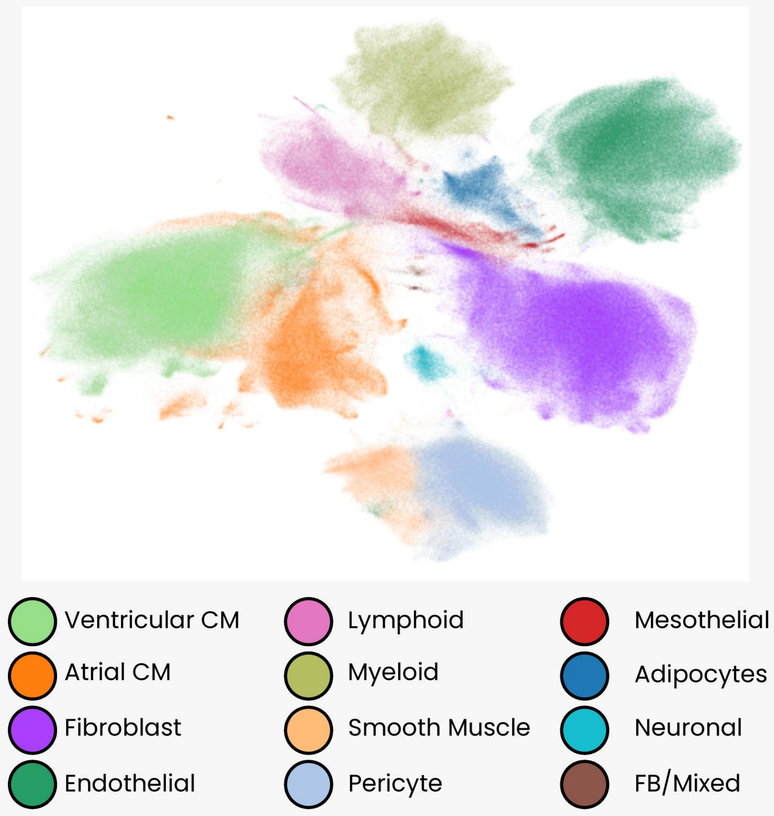
Single nuclear sequencing can capture more cardiomyocytes, but we can also get a bigger population of CMs to study by integrating multiple single cell datasets. Combining public datasets can enable new comparisons to be made between diseases, heart regions, or demographic groups that weren't present within individual studies.

USING GALAXY

Seven selected datasets were integrated and analysed using the Single Cell Lab on the Galaxy EU server. In addition to providing single cell tools and free computing resources, Galaxy provided several features that helped keep the data and results organised.

CELL TYPE ANNOTATIONS

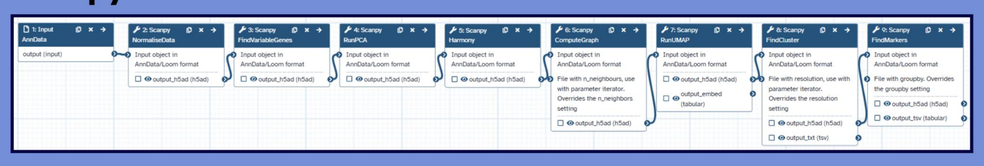
UMAP of the combined dataset (Scanpy pipeline)



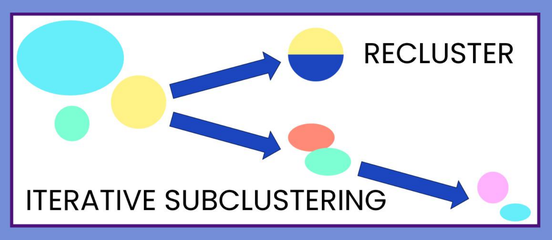
SUBCLUSTERING WORKFLOWS

Two healthy reference datasets of the adult human heart were combined with four datasets that included donors with AF or Heart Failure (HF). Cardiomyocytes from a foetal dataset were also included to determine whether the diseased cells showed any signs of dedifferentiation, which has previously been suggested to occur in AF. Both the Scanpy and Seurat pipelines were used to cluster this combined dataset.

Scanpy



Seurat

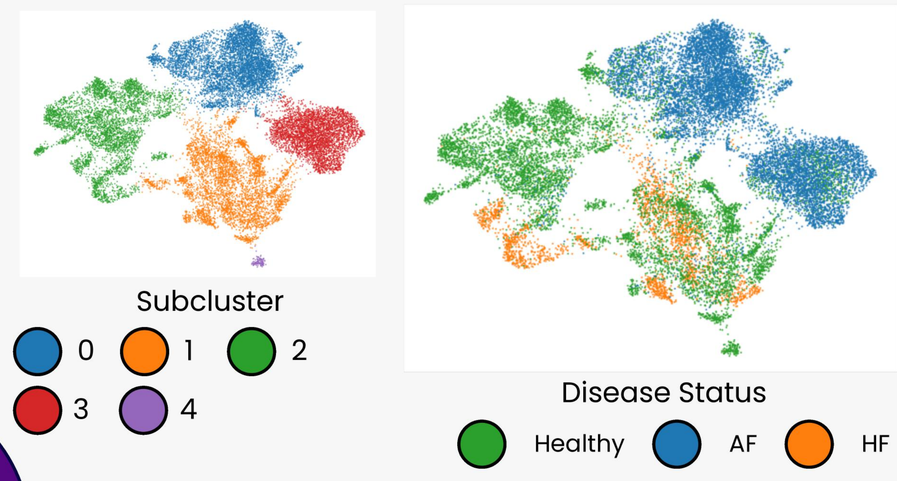


After the initial round of clustering to identify cell types using canonical markers, the cardiomyocytes were subclustered. Rather than simply changing the resolution of the CM clusters or re-clustering using the same Highly Variable Genes (HVGs), the entire clustering workflow was rerun to reveal differences within this cell type.

THE SEURAT TOOLS WERE WRAPPED DURING AN ELIXIR STAFF EXCHANGE WITH THE GALAXY EU TEAM

CM SUBCLUSTERS

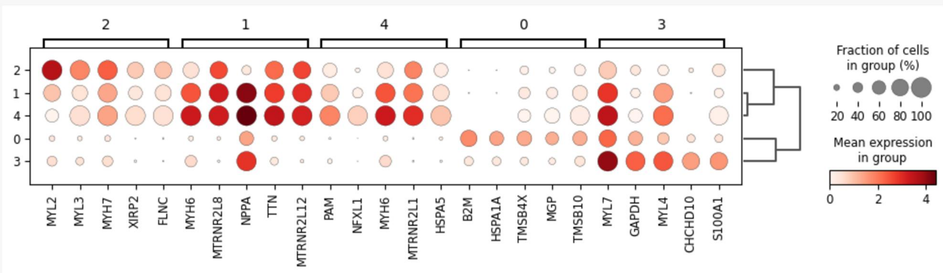
UMAPs of the adult CM subclusters for scRNA data only (Scanpy pipeline)



HOW DO CMs CHANGE IN AF?

Cardiomyocytes from healthy donors typically form separate clusters of atrial and ventricular cells. The atrial and ventricular CM clusters for the combined dataset were not as clearly differentiated and appear intermingled in the top UMAP. During subclustering, the diseased atrial CMs sometimes clustered with the ventricular CMs. When CMs from a Foetal scRNA dataset were included, a subset of the diseased CMs clustered with these developing cells instead of the adult CMs. Subcluster 0 in the adult scRNA UMAP shown to the right mainly contained CMs from donors with AF. This cluster expressed genes that aren't commonly seen in adult CMs, but which have been suggested to play roles in dedifferentiation and responses to cell stress or injury.

Dotplot of the top differentially expressed markers for the adult CM scRNA subclusters



Meeting Attendance supported by BSCB British Society for Cell Biology, The Company of Biologists

INTEGRATED DATASETS

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